ROMAN CHAKRABORTY

Full-Stack Developer | AI/ML Developer (RAG & LLM Applications)
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PROFESSIONAL SUMMARY

Full-Stack and AI developer (MCA, 2026) who builds end-to-end web applications with React, Next.js, Node.js, Express.js, and FastAPI, and ships LLM features such as RAG, semantic search, and multimodel pipelines using LangChain, LangGraph, and Qdrant. Skilled in REST API design, MongoDB, MySQL, Docker, and JWT authentication.

SKILLS

Languages: Python, JavaScript, HTML, CSS
Framework: Next.js
Frontend: React.js, Vite, Tailwind CSS, Bootstrap
Backend & Databases: Node.js, Express.js, FastAPI, REST APIs, JWT, MVC | MongoDB, MySQL, Qdrant (Vector Database)
AI & Machine Learning: RAG, LangChain, LangGraph, LLMs, Vector Embeddings, Gemini API, Grok API
Tools: Docker, Git, GitHub, Postman, Cloudinary, Claude Code

PROJECTS

YojanaAI - AI-Powered Government Scheme Discovery Platform | **GitHub** | June 2026 - September 2026
Next.js, FastAPI, Python, MongoDB, Qdrant, LangChain, LangGraph, Docker, REST APIs, Grok API

- Built **YojanaAI, an AI-powered government scheme discovery platform** that helps users find relevant Indian government schemes based on their **needs**, **eligibility**, and **natural-language queries** across **4,900+ schemes**.
- Engineered an end-to-end RAG system that converts scheme data into **24,850+ vector embeddings** in Qdrant and uses semantic retrieval with MMR to identify diverse, contextually relevant schemes instead of relying on keyword matching.
- Built a **multimodel AI pipeline** integrating **3 LLMs** with LangChain and LangGraph, and developed the full-stack platform with Next.js, FastAPI, MongoDB, Qdrant, Docker, and REST APIs for eligibility filtering, scheme details, official application links, and AI-powered recommendations.

RAgeniusAI - AI-Powered PDF Learning Assistant | **GitHub** | April 2026 - June 2026
React.js, FastAPI, Python, MongoDB, Qdrant, LangChain, Hugging Face, Gemini API, Grok API, Tailwind CSS

- Built a RAG pipeline for conversational PDF search, chunking documents and indexing embeddings in Qdrant, cutting re-reading of **10 to 100+** page documents down to a single query.
- Tested retrieval across **15+** PDFs and refined chunking to improve retrieval on multi-column documents.
- Generated **summaries** and **notes** with **Gemini** and built **quiz** and **PDF Q&A** features with **Grok.**

Davis-Avery - Full-Stack E-Commerce Platform | **GitHub** | September 2025 - December 2025
React.js, Node.js, Express.js, MongoDB, Razorpay, JWT, Cloudinary, Tailwind CSS

- Designed the MongoDB schema and REST API for **50+ product** listings using **MVC architecture**, with category and price filtering, search, and browsing.
- Built a shopping cart with size selection, secured with **JWT authentication**, protected routes, and **rate-limited** login/signup.
- Integrated **Razorpay** in test mode (**4** test transactions) with failed/pending payment handling, and built an admin dashboard for inventory and order management with low coupling and high cohesion.

EDUCATION

Master of Computer Application (MCA)
Kalyani Government Engineering College, West Bengal | 2024 - 2026 | **CGPA: 8.26**
Bachelor of Science (B.Sc.) in Physics | Calcutta University, West Bengal | 2020–2023 | **CGPA: 7.87**

ACHIEVEMENTS & CERTIFICATIONS

- **Rank 82 in WBJECA** (West Bengal Joint Entrance Examination for MCA), September 2024
- **Internet of Things (IoT)** - NPTEL (SWAYAM) | **Link**
- **System Analysis & Design** - Kalyani Government Engineering College (KGEC) | **Link**